# A DUALIZING COMPLEX RESTRICTS TO A DUALIZING COMPLEX UNDER LOCALISATION

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- The statement that localisation of a bounded complex of injectives stays a bounded complex of injectives over the localised ring. Invoked inside the counted proof (source lines 2958-2961). Gives finite injective dimension of the localised complex.
- The base change statement for internal homs along flat maps. Invoked at the end of the counted proof (source lines 2969-2973) as a cross-chapter pointer. Identifies the double dual of the localised complex with the localised ring.

*Source reference.* Original lemma lemma-localization-injective-modules: sources/stacks-dualizing/pinned-original-source.tex, source lines 527–527. Navigation pointer only: the counted text refers to the Stacks item lemma-localization-injective-modules, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original reference more-algebra-lemma-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2719–2719. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

**Lemma 0.1.** *Let $A$ be a Noetherian ring. Let $B = S^{-1}A$ be a localization. If $\omega_A^\bullet$ is a dualizing complex, then $\omega_A^\bullet \otimes_A B$ is a dualizing complex for $B$.*

*Date*: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Let $\omega_A^\bullet \to I^\bullet$ be a quasi-isomorphism with $I^\bullet$ a bounded complex of injectives. Then $S^{-1}I^\bullet$ is a bounded complex of injective $B = S^{-1}A$-modules (Lemma lemma-localization-injective-modules) representing $\omega_A^\bullet \otimes_A B$. Thus $\omega_A^\bullet \otimes_A B$ has finite injective dimension. Since $H^i(\omega_A^\bullet \otimes_A B) = H^i(\omega_A^\bullet) \otimes_A B$ by flatness of $A \to B$ we see that $\omega_A^\bullet \otimes_A B$ has finite cohomology modules. Finally, the map

$$B \longrightarrow R\mathrm{Hom}_A(\omega_A^\bullet \otimes_A B, \omega_A^\bullet \otimes_A B)$$

is a quasi-isomorphism as formation of internal hom commutes with flat base change in this case, see More on Algebra, Lemma more-algebra-lemma-base-change-RHom. □